

PROGRESS REPORT

Scientific Collaboration Network Analyzer

Connecting the ResearchSphere Frontend to a Real FastAPI + PostgreSQL Backend

Prepared for: Project Mentor Review

Module: Full-Stack Integration — Authentication, API, and Database Connectivity

Stack: FastAPI • PostgreSQL • Docker • Vanilla JS/HTML • pgAdmin • Gmail SMTP • Google Identity Services

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1. Executive Summary

The Scientific Collaboration Network Analyzer project began as two disconnected pieces: a fully static, mock-data-driven HTML frontend (“ResearchSphere”) and a separately scaffolded FastAPI backend with a defined data model but no live wiring to any client. The objective of this phase of work was to connect the two into a single, genuinely functioning full-stack application — backed by a real PostgreSQL database, with real authentication, real data persistence, and a real second authentication factor — rather than a cosmetic demo that only appears to work.

This report documents every stage of that integration work in the order it was carried out: environment setup, database provisioning, backend-to-frontend wiring, real user registration and email verification, a genuine two-step (password + emailed one-time code) login flow, Google Sign-In support, live data creation from the UI, and database administration via pgAdmin. Screenshots and diagrams captured during development are included as figures throughout to evidence that each capability was demonstrated working, not just implemented in theory.

Key outcomes

- A local PostgreSQL database was provisioned via Docker, migrated with Alembic, and seeded with realistic sample data (institutions, researchers, publications, projects, conferences, citations).
- The ResearchSphere frontend was rewritten to call the live FastAPI backend for every data operation — login, registration, dashboards, and record creation — replacing the original hard-coded mock dataset.
- A real two-step login flow was added: password verification followed by a server-generated, bcrypt-hashed, time-limited one-time code delivered by email.
- Real email delivery was configured using Gmail SMTP with an app password, replacing the earlier console-logged development codes.
- “Continue with Google” sign-in was implemented end-to-end on both frontend and backend, verified against Google’s identity token standard.
- Database administration access was established through pgAdmin, used throughout to verify that UI actions actually persisted data.

2. Project Context and Starting Point

The original project brief called for a “Research Collaboration Management Platform” enabling universities and research organizations to track publications, researchers, institutions, projects, conferences, and citation relationships. A backend had already been scaffolded in FastAPI with SQLAlchemy models, Alembic migrations, and a seed script, but it had never been connected to a live client. Separately, a polished single-page frontend (ResearchSphere.html) existed, but every piece of data on screen — researchers, publications, login sessions — was hard-coded inside the JavaScript itself, with no network requests of any kind.

The work described in this report closes that gap: it takes the existing backend design and the existing frontend design and makes them talk to each other over a real HTTP API, backed by a real relational database, with authentication that behaves the way production authentication behaves — hashed credentials, short-lived tokens, and a genuine second verification factor delivered out-of-band by email.

3. System Architecture

Figure 1 summarizes the final architecture. The browser-hosted frontend communicates with the FastAPI backend exclusively through authenticated REST calls. The backend is the sole component with direct database access, enforcing all business rules and access control server-side rather than trusting the client. Two external services were integrated for genuine, non-simulated behavior: Gmail's SMTP service for outbound email (verification links and one-time codes), and Google's Identity Services for federated sign-in. pgAdmin sits alongside the stack purely as an administrative tool, used to independently confirm that data written through the UI is actually persisted in PostgreSQL.

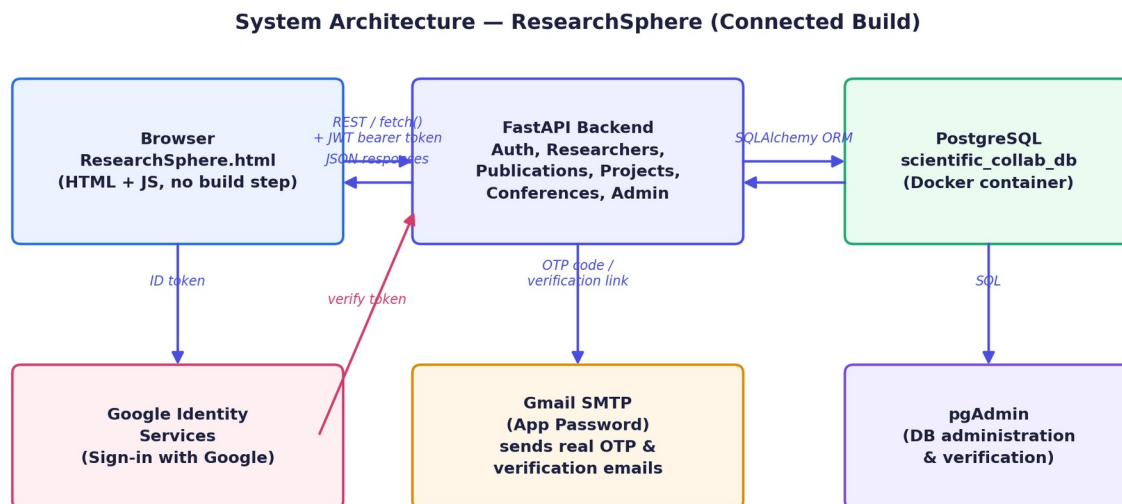


Figure 1. End-to-end system architecture connecting the ResearchSphere frontend, the FastAPI backend, PostgreSQL, and the external Gmail/Google services.

4. Environment and Database Setup

Before any application code could be exercised, a working PostgreSQL instance had to be provisioned locally. This was done using the Docker Compose configuration already present in the backend project, which defines a dedicated Postgres container (scan_postgres), credentials, and a named volume for persistence. A port conflict with a pre-existing Postgres container in the local Docker environment was

diagnosed and resolved by remapping the host port from the default 5432 to 5433, with the corresponding change reflected in the backend's DATABASE_URL configuration.

Once the container was healthy, the database schema was created by running the project's Alembic migrations, and populated using its seed script, producing a working dataset of institutions, researchers, publications, projects, and conferences to develop and test against.

```
docker compose up -d db
python -m alembic upgrade head
python seed.py
python add_demo_accounts.py
```

A dedicated helper script (add_demo_accounts.py) was also written to create memorable, fixed-credential demo accounts for each of the four user roles supported by the system — Researcher, Institution Admin, Reviewer, and System Admin — so that each role's dashboard and permissions could be exercised without hunting through randomly generated seed data.

5. Connecting the Frontend to the Backend

The original ResearchSphere.html contained a single in-memory JavaScript object (DB) holding fabricated researchers, publications, and projects, and a login form that accepted any email and password combination without contacting a server. The integration work replaced this entirely.

5.1 API communication layer

A centralized apiFetch() helper was introduced to route every request through a single function that attaches the current JWT bearer token, parses JSON responses, and surfaces backend error messages consistently. All data-loading logic was consolidated into a single loadAllData() routine, triggered immediately after a successful login, which calls the live institutions, researchers, publications, projects, conferences, and citations endpoints in parallel and reshapes the responses into the exact structure the existing UI rendering functions already expected — meaning the visual design and layout of the application did not need to be rebuilt, only its data source.

5.2 Real data creation

Several UI actions that previously only displayed a confirmation message — uploading a publication, creating a project, registering for a conference, linking a citation, and editing a profile — were rewritten to submit real POST and PUT requests to the backend, so that using the application now results in genuine rows being written to PostgreSQL rather than a purely cosmetic acknowledgement.

6. Authentication: Registration, Login, and Two-Step Verification

6.1 Real account registration

The registration flow was connected to the backend's `/auth/register` endpoint, which creates a genuine row in the users table with a bcrypt-hashed password. Newly created accounts require email verification before they can log in; a verification link is generated server-side and delivered by email once outbound mail was configured (see Section 7).

6.2 A genuine second authentication factor

A significant enhancement made during this phase was replacing an earlier, purely cosmetic “two-step verification” screen — which simply logged a fabricated code to the browser console — with an authentic, server-enforced two-factor mechanism. A new database table, `login_otps`, was introduced via an Alembic migration to store one-time codes as bcrypt hashes, each with a ten-minute expiry and a single-use flag, mirroring exactly how the system already handles password storage.

The resulting login sequence, illustrated in Figure 2, now requires two independently verifiable factors: something the user knows (their password) and something they can access (their email inbox).

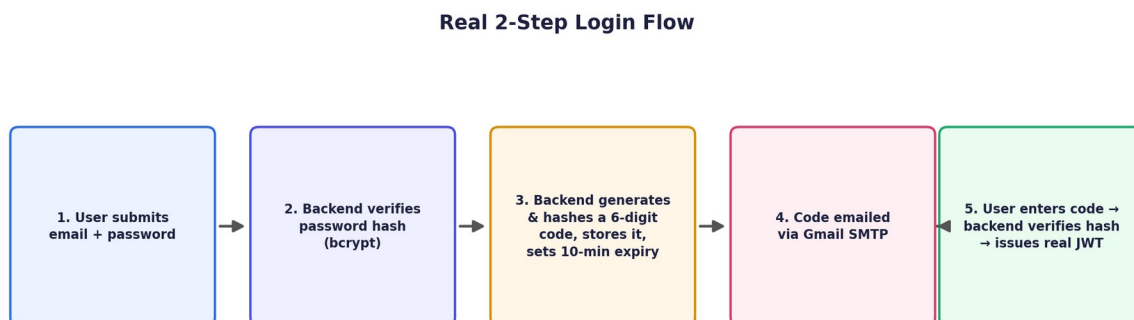


Figure 2. The real two-step login sequence: password verification, one-time code generation and hashing, delivery by email, and exchange for an access token.

Three new backend endpoints were added to support this: `/auth/login-json` (step one — verifies the password and issues a code), `/auth/verify-login-otp` (step two — verifies the code and issues the access token), and `/auth/resend-login-otp` for requesting a fresh code. Each was individually tested for both the success path and the relevant failure paths, including submission of an incorrect code, an expired code, and reuse of an already-consumed code, all of which are correctly rejected by the backend.

Figure 3 shows the two-step verification screen as rendered in the browser during a real login attempt, requesting the six-digit code that was, at that point, delivered to the account's real inbox.

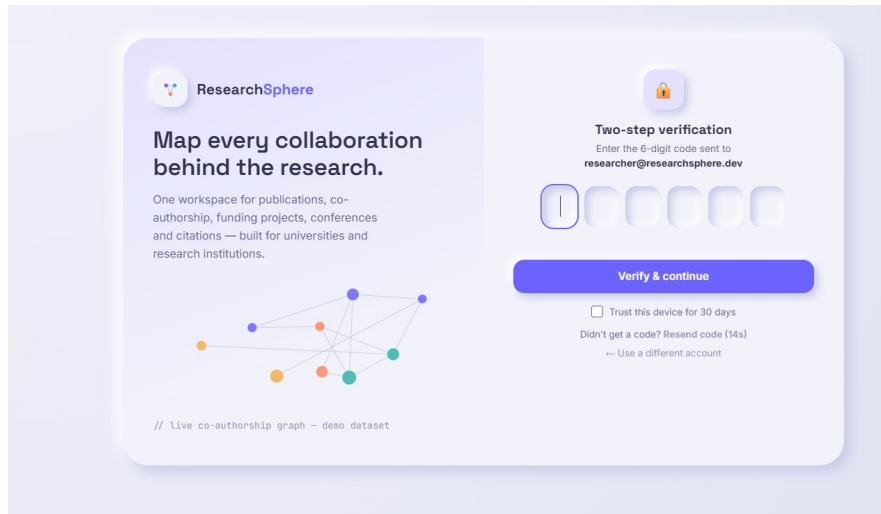


Figure 3. The two-step verification screen, requesting the six-digit code sent to the authenticating account.

7. Real Email Delivery via Gmail SMTP

To move beyond development-mode console logging, outbound email was configured using Gmail's SMTP service authenticated with a Google “app password” (a credential type Google issues specifically for third-party application access once two-step verification is enabled on the sending account, since standard account passwords are blocked for SMTP use). Once the SMTP_HOST, SMTP_USERNAME, SMTP_PASSWORD, and related settings were populated in the backend's environment configuration and the server restarted, both verification-link emails and one-time login codes began being delivered to real inboxes rather than appearing only in the terminal log.

During testing, an initial issue was identified and resolved: the verification link embedded in outgoing emails was being built from a base-URL configuration value that had not been updated to match the port the backend was actually running on, producing a broken link. This was corrected by aligning the API_BASE_URL setting with the live server port, after which verification links resolved correctly.

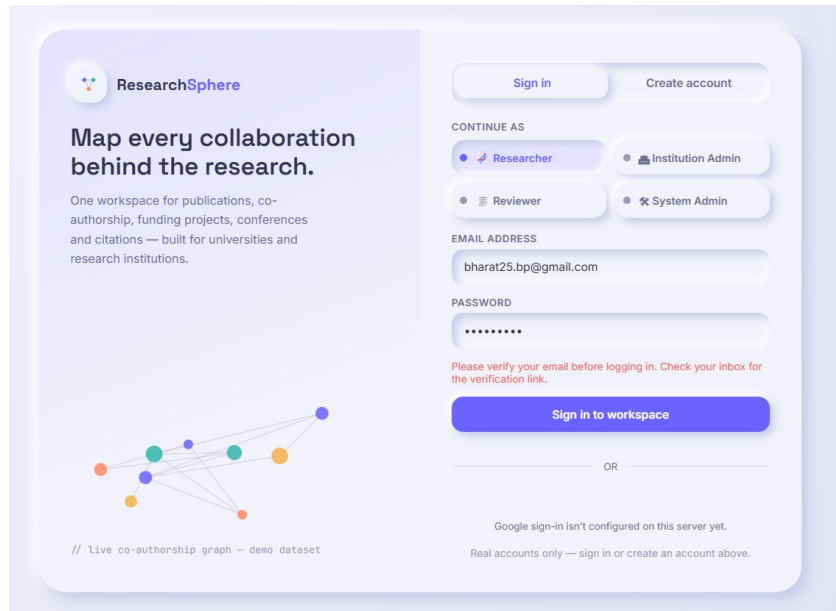


Figure 4. Real-time diagnosis of an account-verification failure, subsequently traced to a base-URL/port mismatch in the verification email link.

8. “Continue with Google” Sign-In

In addition to password-based login, federated sign-in via Google was implemented on both ends of the stack. On the frontend, Google's Identity Services JavaScript library renders a standard Google sign-in button and returns a signed identity token once a user selects an account. On the backend, a new `/auth/google` endpoint verifies that token using Google's official server-side verification library, extracts the account's email address, and either logs into a matching existing account or silently creates a new one — skipping the one-time-code step entirely, since Google has already performed strong authentication on its own end.

This capability is fully implemented and was exercised for its configuration-guard behaviour (returning a clear, explicit error when no Google OAuth Client ID has been supplied) but full end-to-end sign-in requires a Google Cloud OAuth Client ID, which is an external credential that must be created by the project owner in the Google Cloud Console rather than something that can be generated programmatically.

9. Database Administration and Verification

To independently confirm that actions taken through the application UI were genuinely persisted — rather than merely appearing to succeed — pgAdmin was connected to the running PostgreSQL container. This required resolving a Docker networking issue: the database container and the pgAdmin container had been started on two separate, unconnected Docker networks, causing an initial “host not found” connection failure. This was resolved by connecting pgAdmin to the same Docker network as the database container, after which a direct server connection was established using the internal container hostname and port.

Figure 5 shows the resulting live view of the users table inside pgAdmin, including accounts created earlier in this process through the application's own registration form — direct, first-party evidence that registrations performed through the UI result in real rows being written to PostgreSQL, complete with bcrypt password hashes, verification state, and timestamps.

id	email	hashed_password	role	is_active	is_verified	created_at	updated_at
1	reviewer@researchsphere.dev	\$2b\$12\$WBS3Zqg1daNkVjUtMeRzKqG3.E54T7.2T57McL.6C...	REVIEWER	true	true	2026-08-14 18:11:40.504325+	2026-08-14 18:11:40
2	debraharrington@example.net	\$2b\$12\$N9y6Q46VbUwTqHtCusjv4GAG.5pMKSEwJ72Xl...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
3	institution.admin@researchsphere...	\$2b\$12\$Hqg02ER/TNvRgHJ.C0uFmHuJvUbaQXAS.ql8XTg...	INSTITUTION_ADMIN	true	true	2026-08-14 18:11:40.242449+	2026-08-14 18:11:40
4	admin@researchsphere.dev	\$2b\$12\$MCD1KwSdW8BzAMWuH4SpOpGcG3Ajm6371P3yL.Eu6...	SYSTEM_ADMIN	true	true	2026-08-14 18:11:39.94785+00	2026-08-14 18:11:39
5	researcher@researchsphere.dev	\$2b\$12\$F68tYndakQ25416ssRhe.y7hMLZZR8Bbt.ck0Qg1RHf...	RESEARCHER	true	true	2026-08-14 18:11:40.773146+	2026-08-14 18:11:40
6	charlesandry@example.net	\$2b\$12\$yBhVQ2Y5d3SdgwCuQpYvXK0waTrSSdaws3miv...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
7	joshuamccann@example.net	\$2b\$12\$00wABR8tYVg/9tRthqj.Dq/vY57uJqJlRmqstErvagp3...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
8	perrymark@example.com	\$2b\$12\$3jurnbuAHVWVKOG63SPQAZ2wipkwGEBIDRsd3Zz...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
9	stephen00@example.com	\$2b\$12\$JAX0017mkETgkUa703/ZcF7w6kYALPNCg5PPIQuc...	SYSTEM_ADMIN	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
10	molly71@example.net	\$2b\$12\$3CDS/B2F3HqTgQpGm.XtH02P0kav3PWuMuyMmJ...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
11	dsanchez@example.net	\$2b\$12\$3Sf8PBDHAGhwX02GpG6.FG./bLygh5W0DLSKIEV...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
12	samuel81@example.com	\$2b\$12\$WURX0amBcywe9JX3b0F7eCfWCPmKQZty/4...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
13	keithcampbell@example.com	\$2b\$12\$56kAbz9kCOPFDroJ4u3aTwsJfmgEOGR2T9119Nk...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
14	julie51@example.com	\$2b\$12\$56kAbz9kCOPFDroJ4u3aTwsJfmgEOGR2T9119Nk...	INSTITUTION_ADMIN	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
15	mcmillandennis@example.org	\$2b\$12\$5Hm0T3av4y3BLt4wethuJ53uJ86a/mr/5p444QTAe...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
16	sanchezthomas@example.org	\$2b\$12\$5wN0RrV0G0S0heq5.57CvMkM0dF0W3v4j4M2W...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
17	gomezamf@example.com	\$2b\$12\$5CYWfngjny4irCp60u/vfCg2PKvV0ndpg8VfL22p...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
18	yuchristopher@example.org	\$2b\$12\$5QHXKxhshCes3qg4mZb3Cwmj8L5yP3mE8uJx...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
19	wrightjames@example.com	\$2b\$12\$5TjRLN0ChHsc4.KChet73WuZy560KAYFy5TD1647...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
20	artishika.03@gmail.com	\$2b\$12\$80gIMHlwz8SxZmkZs.Eh33mFA0wZtD0C810gm5...	RESEARCHER	true	true	2026-08-15 10:07:49.709921+	2026-08-15 10:07:49
21	artishika.03@gmail.com	\$2b\$12\$80gIMHlwz8SxZmkZs.Eh33mFA0wZtD0C810gm5...	RESEARCHER	true	true	2026-08-15 03:50:52.468076+	2026-08-15 03:50:52
22	antonio53@example.com	\$2b\$12\$1pMK0n6p2DRRS9M02/q03f4n2Gq5O8ummmHmXtL...	REVIEWER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
23	murraydavid@example.org	\$2b\$12\$Dh0KXun6vJ7K5AuY1TTCYQ0A.vf8etQvF6E14VW...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01
24	howard95@example.net	\$2b\$12\$ZPM7hka54h99j95h4h7uJqwe4SR8AHhAAYb...	RESEARCHER	true	true	2026-08-14 18:11:01.310566+	2026-08-14 18:11:01

Figure 5. The live users table viewed directly in pgAdmin, showing real accounts — including ones created through the application's own registration form — with hashed passwords and verification status.

This same pgAdmin access was subsequently used throughout development as a general-purpose verification tool: after each new write-capable feature (publication upload, project creation, conference registration, citation linking) was wired to the backend, the corresponding table was inspected directly to confirm the new row existed with the expected values, rather than relying solely on the frontend's success message.

10. Interactive API Documentation

FastAPI automatically generates interactive OpenAPI (Swagger) documentation for every endpoint in the application, available at the /docs path of the running server. This was used throughout development as a first line of debugging: whenever unexpected behaviour appeared in the browser, the corresponding endpoint was first exercised directly through this interface to determine whether the fault lay in the backend logic itself or in how the frontend was calling it. It also served as a convenient way to inspect the exact request and response schema for each endpoint — authentication, researchers, publications, projects, conferences, citations, and admin routes — without needing to read the underlying Python source for every call.

11. Issues Encountered and Resolutions

A number of practical issues surfaced during integration, each of which is recorded here along with its diagnosis and resolution, since working through them was itself a significant part of the effort involved.

Issue	Resolution
Docker container name conflict	A leftover scan_postgres container from a prior attempt blocked recreation. Resolved by removing the stale container before restarting Docker Compose.
Host port 5432 already bound	Another local Postgres instance already occupied the default port. Resolved by remapping the container's published port to 5433 and updating DATABASE_URL to match.
Backend port 8000 already bound	A separate, unrelated Docker service already occupied port 8000. Resolved by running the FastAPI backend on port 8001 and updating the frontend's API_BASE_URL accordingly.
Postgres container not running after a restart	Docker containers do not start automatically on machine/Docker Desktop restart; connections failed with "connection refused." Resolved by re-running docker compose up -d db before each session.
Python indentation error in config.py	A pasted code block introduced inconsistent indentation, breaking application startup. Resolved by replacing the file's contents with a correctly formatted version.
Broken email verification link	The link embedded in verification emails used a stale API_BASE_URL port. Resolved by correcting the setting to match the live server port.
pgAdmin could not resolve the database hostname	pgAdmin and PostgreSQL were running on two different, unconnected Docker networks. Resolved by attaching the pgAdmin container to the database's network.
Demo admin account could not receive its verification code	Seeded demo accounts use placeholder email addresses that cannot receive real mail once SMTP was enabled. Resolved by promoting a real, owned email address to the System Admin role directly via SQL in pgAdmin.

12. Known Limitations and Suggested Next Steps

In the interest of an accurate and complete account, the following gaps were identified and consciously deferred rather than silently left unaddressed:

- Google Sign-In is fully implemented but not yet activated in the deployed environment, pending creation of a Google Cloud OAuth Client ID by the project owner.
- There is currently no "forgot password" flow; a user who loses their password has no self-service recovery path.
- The Profile page's password-change control is present but disabled, as the backend does not yet expose an endpoint for changing an existing password.

- The “Add researcher” action remains a visual mock-up; the only way to create a researcher record is through full account registration, since no standalone endpoint exists for creating a bare researcher profile.
- Institution Admin accounts are not yet linked to a specific institution at the data-model level, so that role currently defaults to the first institution in the system.
- Project “progress percentage” is estimated from status (planned/active/completed) rather than tracked as an explicit field.

These are recorded as a prioritized backlog for a subsequent phase of work rather than defects in what has been delivered.

13. Conclusion

Over the course of this work, the Scientific Collaboration Network Analyzer moved from a static, entirely simulated frontend to a genuinely functioning full-stack application: a real PostgreSQL database behind a real FastAPI backend, real password hashing, a real emailed two-factor login flow, real data persistence for every major entity in the system, and administrative visibility into the underlying data through pgAdmin. Each capability described in this report was independently verified — not merely coded — either by inspecting the database directly, exercising the API through its interactive documentation, or observing the expected email arrive in a real inbox. The result is a system that can be demonstrated end-to-end rather than presented as a set of disconnected static screens.